

Unit Testing

- Testing specific units of code independently.

Ex.

```
class Calculator {
```

```
    Add() {
```

```
        // logic
    }
```

```
    sub() {
```

```
        // logic
    }
```

testing Add()
independently
is called unit
testing

If we are testing all the methods together then call it as integration testing

Advantages of Unit testing

- Find bugs early
- Easy bugs fixing
- Reduces cost & time

How to perform unit test

JUnit

→ Most popular framework for unit testing

→ Imp versions

- JUnit - 4
- JUnit - 5

What is Mockito?

→ It is a mocking framework

Mocking → Creating something fake

ex.: Let's suppose you have created a depd and you have to interact with a database but you don't have any in reality so in such cases we use Mockito to create a fake DB just to check business logic

Imp Annotations of JUnit 4

@Test → Indicates that the below method is test cases

@BeforeClass → Executes method before all test cases

@AfterClass → Executes method after all test cases

@Before → Executes method before each test case.

@After → Executes method after each test case.

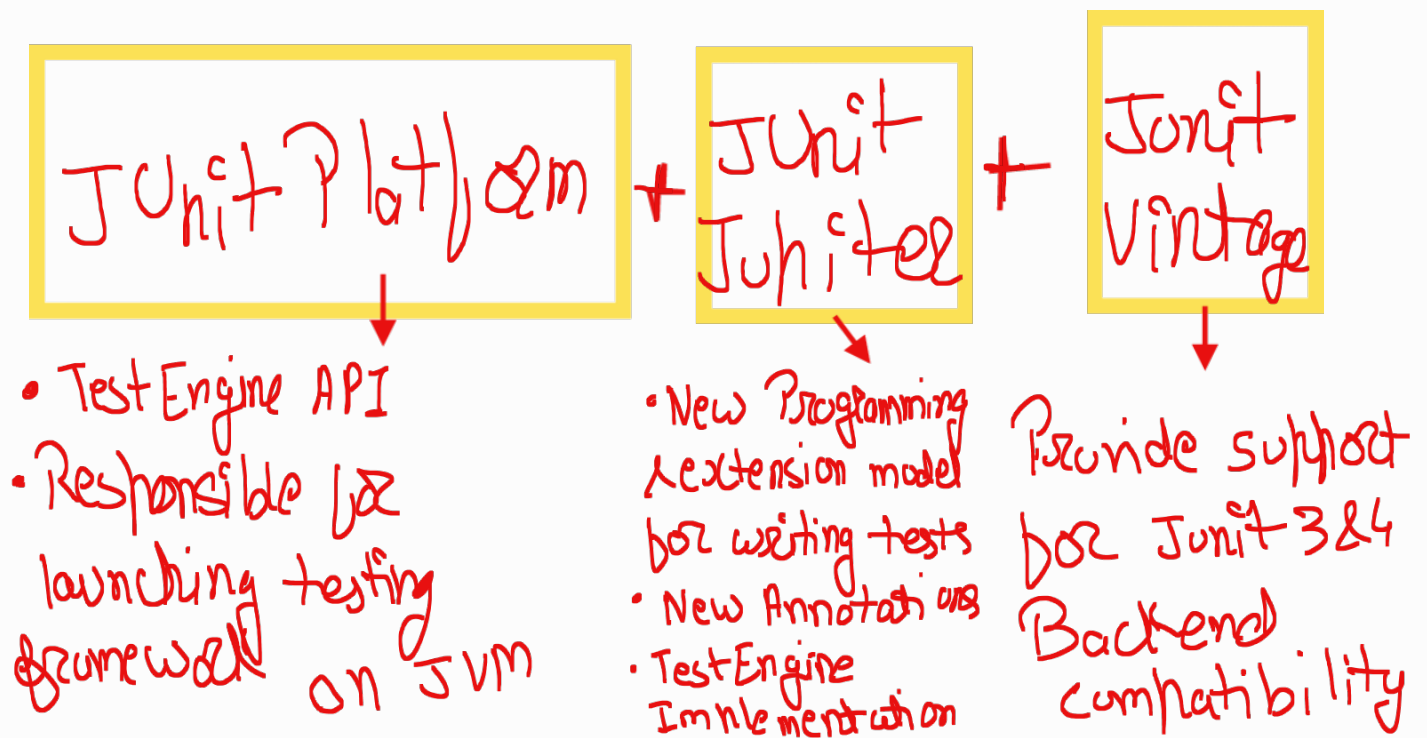
NOTE:

@Test (timeout = 2000) → 2000ms

If test case takes more than 2k ms then it will sleep down

JUnit 5

JUnit 5 is composed of several different modules



- @Test annotation comes from
org.junit.jupiter.api.Test
in JUnit 5

Some common test cases are

@Test :- — 1 —

@BeforeAll :- Runs before other case only one time

@AfterAll :- Runs after all other cases one time only

@BeforeEach :- — 1 —

@AfterEach :- — 1 —

@DisplayName :- Use to display a custom test name

@Disabled :- It ignores a test case

@Tag: These are used to find or discover the class/method.

@Nested: Used to signal that the annotated class is nested, non-static test class that can share setup & state with an instance of its enclosing class

@TestFactory: Used to mark that the annotated method is testfactory method.

This annotation is factory for test cases

What is Assertions

→ Validating expected o/p will
actual one is called assertions